

Final Project: an Exploration of the ‘airquality’ Dataset

2025-10-19

Introduction

‘airquality’ is a built-in data set in R that was compiled from archives of the New York State Department of Conservation and the National Weather Service, which records the daily air quality measurements in New York City from May 1 to September 30, 1973. Besides Day and Month (which are represented by numeric values), there are four main variables of interest: Ozone, Solar.R, Wind, and Temp. Per R documentation, these variable are defined as the following:

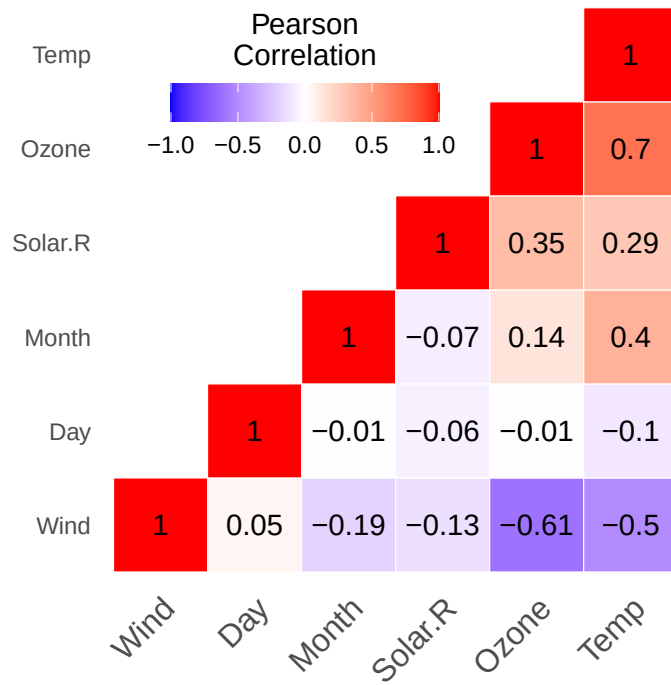
- Ozone: Mean ozone in parts per billion from 1300 to 1500 hours at Roosevelt Island
- Solar.R: Solar radiation in Langleys in the frequency band 4000–7700 Angstroms from 0800 to 1200 hours at Central Park
- Wind: Average wind speed in miles per hour at 0700 and 1000 hours at LaGuardia Airport
- Temp: Maximum daily temperature in degrees Fahrenheit at LaGuardia Airport.

With rising temperatures becoming a key concern for many, the scientific motivation behind our inquiry is to better understand how meteorological factors and air quality measurements affect (or simply have a relationship with) daily maximum temperature.

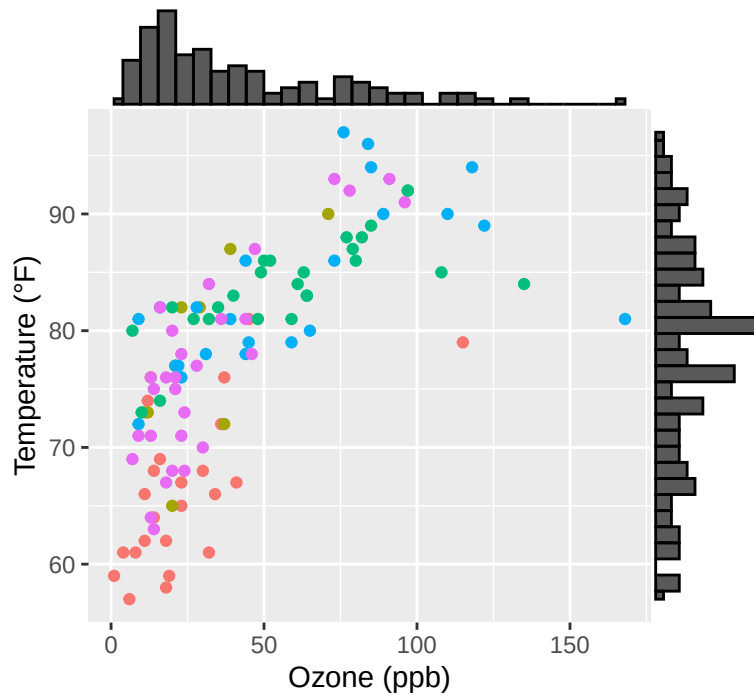
Outcome

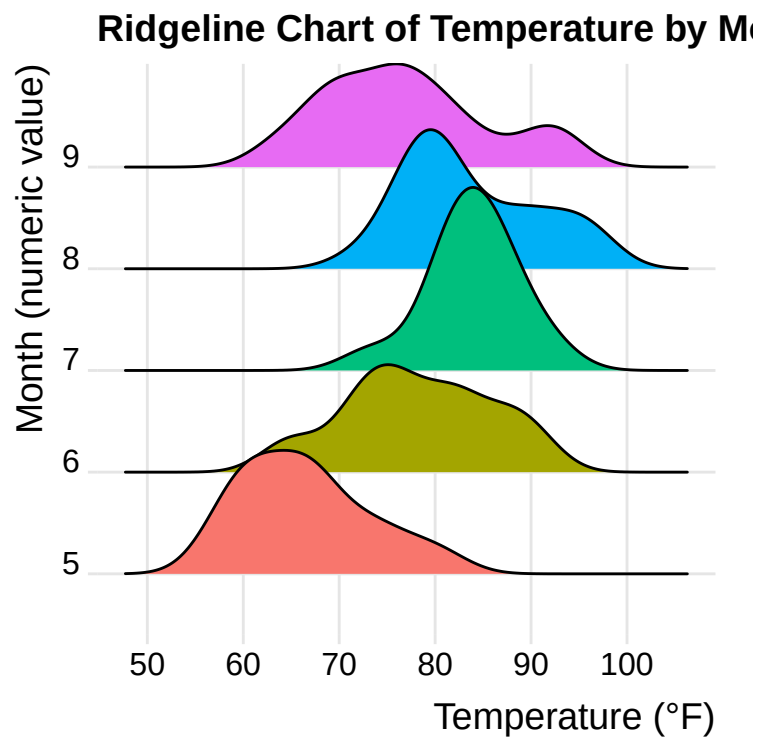
```
## [1] 44
## [1] 153
## [1] 111
```

Air Quality Correlation Matrix



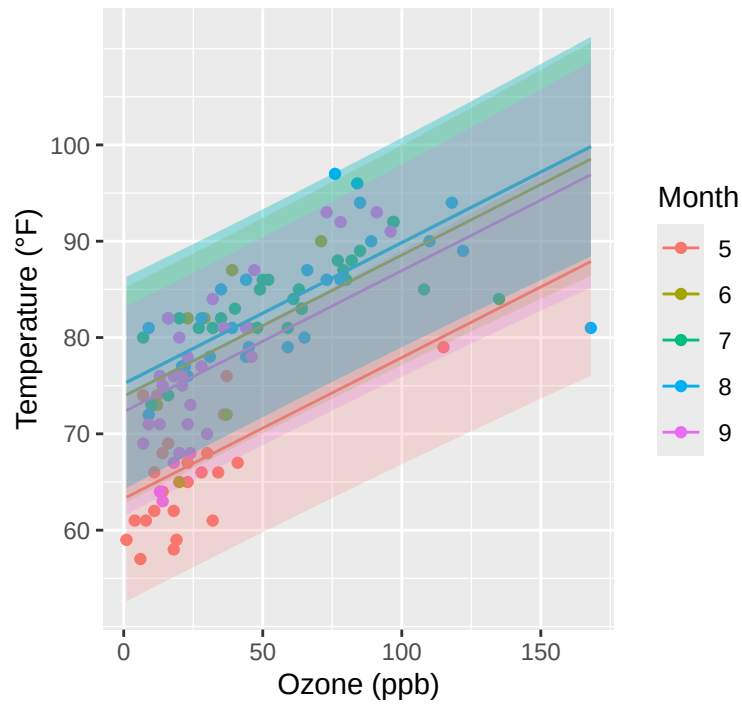
Temperature v. Ozone Scatter Plot w



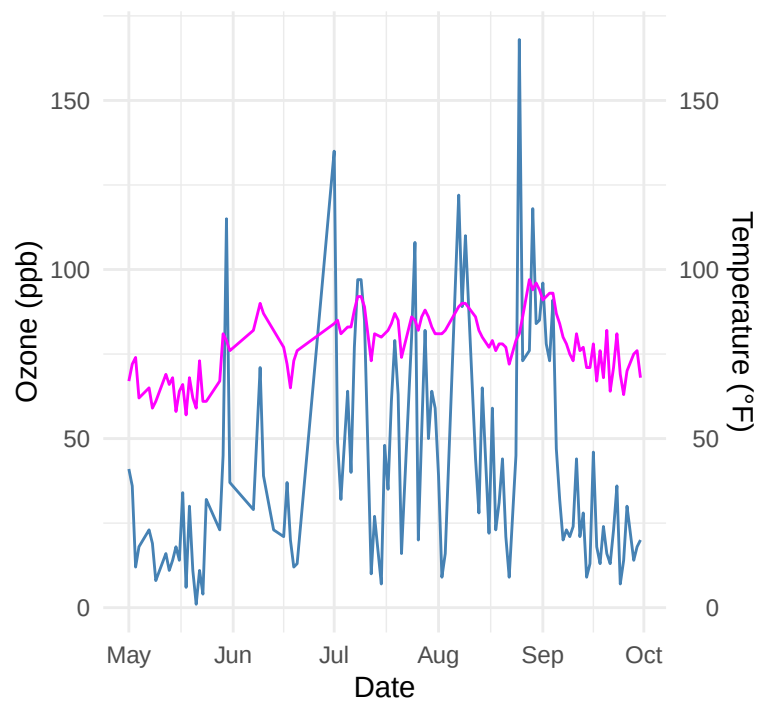


```
##
## Call:
## lm(formula = Temp ~ Ozone + Month, data = new)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.9545  -4.4396   0.5476   4.6118  12.2303
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  55.41339    2.98090   18.589 < 2e-16 ***
## Ozone         0.18735    0.01799   10.412 < 2e-16 ***
## Month         2.00827    0.40636    4.942 2.84e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.215 on 108 degrees of freedom
## Multiple R-squared:  0.5824, Adjusted R-squared:  0.5747
## F-statistic: 75.31 on 2 and 108 DF, p-value: < 2.2e-16
```

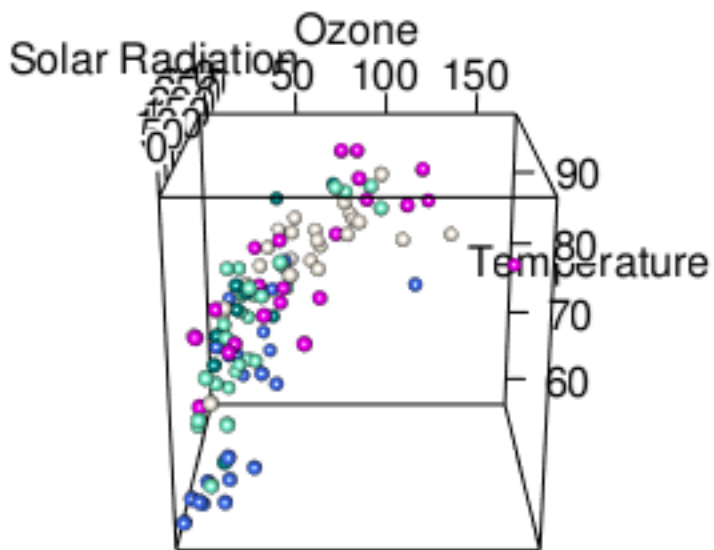
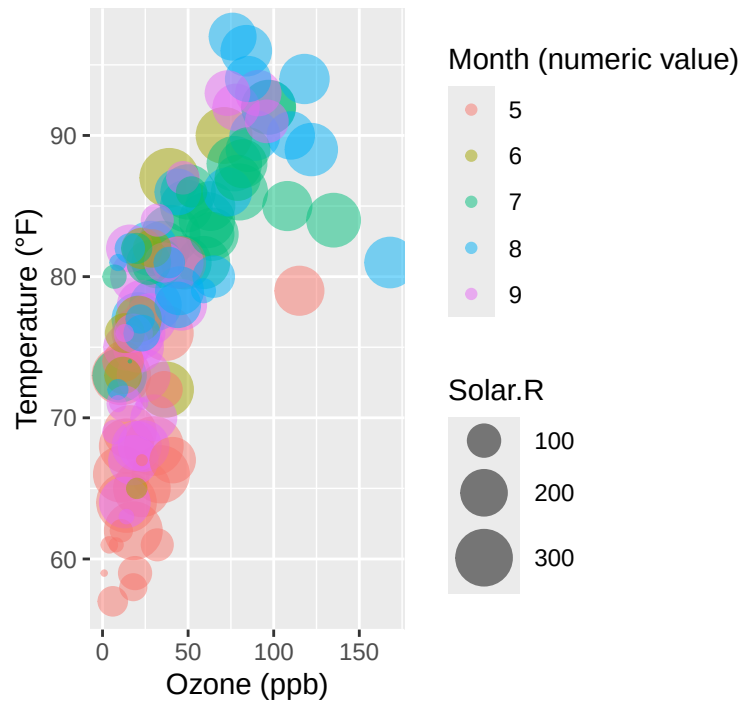
Linear Regression Temperature v. Ozone by

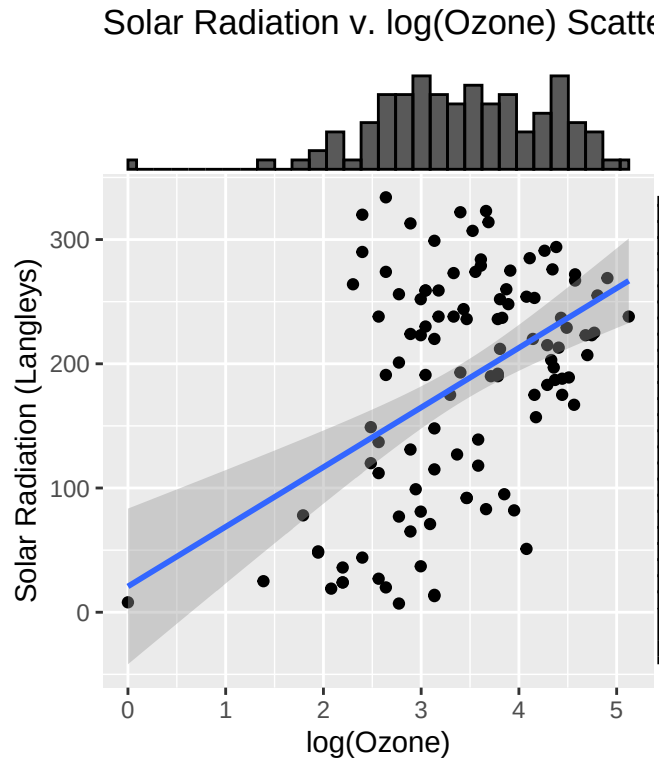


Ozone and Temperature Over Time (1973)



Temperature v. Ozone by Month and Solar F





Discussion

‘airquality’ has 44 NA values. We then eliminated rows with NA values reducing our data set from 153 to 111 rows. From this new data set, the above visualizations were plotted.

In order to explore any potential relationships with our variable of interest, Temperature, we first create a correlation matrix. We find that Ozone has the highest correlation, and thus strongest relationship, with Temperature. With this finding, we then create a scatter plot to explicitly visualize this relationship with each variable’s marginal histograms. The scatter plot hints at Ozone’s positive correlation with Temperature, and our marginal histograms show that Ozone is heavily right skewed while temperature less so.

However, we must take into consideration month of year as this might affect the strength of our relationship since the later summer months tend to have higher maximum daily temperatures due to seasonal change. We illustrate this in our Ridgeline Chart where each distribution of temperature centralizes around higher temperatures with each passing month. Furthermore, we still find that there is a significant linear relationship between Temperature and Ozone when we control for Month. We see that on average for every one-unit increase in Ozone, there is a 0.18735 increase in Temperature.

We, then, plot the linear relationships between Temperature and Ozone, where we find that July and August are hotter than May and June across the Ozone range, and most observed points fall within its ribbons, suggesting the model fits reasonably well. To further illustrate this correlation effect, we plot a times series graph. Temperature and Ozone generally track with each other across time. For example, when there is a spike in Ozone, there is subsequently an increase in Temperature as well.

In addition, we find that both Temperature and Ozone both have a sizeable relationship with Solar Radiation in our correlation matrix. Since higher levels of solar radiation increase the likelihood of forming ground-level ozone, this relationship is expected. Thus, exploring Solar Radiation may aid in our understanding of air quality measurements and their effect on Temperature. We create a bubble plot to help visualize this three-way relationship. However, since there is no noticeable difference in size for an increase across Temperature or Ozone, we decide to better visualize this relationship with a three-dimensional scatter plot of Temperature, Ozone, and Solar Radiation. Visually, we see that there is a weak relationship between Temperature and

Solar Radiation. There is, however, a noticeably stronger relationship with Ozone and Solar Radiation. Interestingly, it takes on a funnel shape, so we graph Solar Radiation v. $\log(\text{Ozone})$ to see if that strengthens its linear relationship.

In order to quantify the strength of the models that we visualized above, we calculate the root mean squared error for each. We found that the RMSE for our model predicting Temperature with Ozone while controlled for month of year was 5.21. However, when we also control for Solar Radiation we find that our RMSE is slightly lower, 5.12. Even smaller yet is when we take the log of Ozone and come out with an RMSE of 5.02.

In all, through this visual and quantitative exploration of the ‘airquality’ data set in R, we found key factors such as Ozone (ppb) and Solar Radiation (Langley) that may influence Temperature ($^{\circ}\text{F}$), which could provide key findings toward understanding and predicting the effects of air quality on rising temperatures by encouraging further interest in how to decrease levels of pollutants that contribute toward smog-forming reactions.

Appendix

```
library(knitr)
library(tidyverse)
library(latex2exp)
knitr::opts_chunk$set(echo = FALSE, message = FALSE, warning = FALSE, fig.width = 4, fig.height = 4, tidy = FALSE)
library(ggplot2)
library(dplyr)
#Clean and explore data
new <- na.omit(airquality)
sum(is.na(airquality))
nrow(airquality)
nrow(new)
# Correlation Matrix
cormat <- round(cor(new),2)
library(reshape2)
melted_cormat <- melt(cormat)
# Get upper triangle of the correlation matrix
get_upper_tri <- function(cormat){
  cormat[lower.tri(cormat)]<- NA
  return(cormat)
}
upper_tri <- get_upper_tri(cormat)
reorder_cormat <- function(cormat){
# Use correlation between variables as distance
dd <- as.dist((1-cormat)/2)
hc <- hclust(dd)
cormat <-cormat[hc$order, hc$order]
}
# Reorder the correlation matrix
cormat <- reorder_cormat(cormat)
upper_tri <- get_upper_tri(cormat)
# Melt the correlation matrix
melted_cormat <- melt(upper_tri, na.rm = TRUE)
# Create a ggheatmap
ggheatmap <- ggplot(melted_cormat, aes(Var2, Var1, fill = value))+
  geom_tile(color = "white")+
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
    midpoint = 0, limit = c(-1,1), space = "Lab",
```

```

    name="Pearson\nCorrelation") +
    theme_minimal()+ # minimal theme
    theme(axis.text.x = element_text(angle = 45, vjust = 1,
        size = 12, hjust = 1))+
    coord_fixed() + labs(title="Air Quality Correlation Matrix")
ggheatmap +
geom_text(aes(Var2, Var1, label = value), color = "black", size = 4) +
theme(
  axis.title.x = element_blank(),
  axis.title.y = element_blank(),
  panel.grid.major = element_blank(),
  panel.border = element_blank(),
  panel.background = element_blank(),
  axis.ticks = element_blank(),
  legend.justification = c(1, 0),
  legend.position = c(0.6, 0.7),
  legend.direction = "horizontal")+
  guides(fill = guide_colorbar(barwidth = 7, barheight = 1,
      title.position = "top", title.hjust = 0.5))
library(ggExtra)

# classic scatter plot :
p <- ggplot(new, aes(x=Ozone, y=Temp, color=factor(Month))) +
  geom_point() +
  theme(legend.position="none") + labs(x="Ozone (ppb)", y="Temperature (°F)", title="Temperature v.

# with marginal histogram
p1 <- ggMarginal(p, type="histogram")
p1
library(ggribes)
# Ridgeline Chart
ggplot(new, aes(x = Temp, y = factor(Month), fill = factor(Month))) +
  geom_density_ridges() +
  theme_ridges() +
  theme(legend.position = "none") +
  labs(x="Temperature (°F)", y="Month (numeric value)", title="Ridgeline Chart of Temperature by Month")
summary(lm(Temp ~ Ozone + Month, data=new))
# Additive Linear Regression
pred_df <- expand.grid(Ozone = (seq(from = min(new$Ozone),
    to = max(new$Ozone),
    length.out = 100)),
    Month = c("5", "6", "7", "8", "9"))
pred_out <- predict(object = lm(Temp ~ Ozone + factor(Month), data=airquality),
    newdata = pred_df,
    interval = "predict")
pred_df <- cbind(pred_df, pred_out)

ggplot(data=pred_df, mapping=aes(x=Ozone, y=fit, ymin=lwr, ymax=upr, fill=factor(Month), color=factor(M
  geom_point(data=airquality, mapping=aes(x=Ozone, y=Temp, color=factor(Month)), inherit.aes = FALSE) +
  scale_y_continuous(breaks=c(60, 70, 80 , 90, 100)) +
  labs(x="Ozone (ppb)", y="Temperature (°F)", color="Month", title="Linear Regression Temperature v. O
  geom_line(data = pred_df, aes(x = Ozone, y = fit)) +
  geom_ribbon(alpha = 0.2, color = FALSE) + guides(fill="none")

```



```

#Predictive Model Evaluation
newdata <- new %>%
  select(-Temp)

pred_out <- predict(object = lm(Temp ~ Ozone + factor(Month), data=airquality),
                    newdata = newdata,
                    interval = "predict")

# Example data
actual <- new$Temp
predicted <- pred_out[, "fit"]

# Compute Mean Squared Error
mse <- mean((actual - predicted)^2)
mse
rmse <- sqrt(mse)
rmse

#Time Series Line Graph
new$Date <- as.Date(with(new, paste(1973, Month, Day, sep="-")))

# Plot Ozone and overlay Temp with secondary axis
ggplot(new, aes(x = Date)) +
  geom_line(aes(y = Ozone), color = "steelblue") +
  geom_line(aes(y = Temp), color = "magenta") +
  scale_y_continuous(
    name = "Ozone (ppb)",
    sec.axis = sec_axis(~ ., name = "Temperature (°F)")
  ) +
  labs(
    title = "Ozone and Temperature Over Time (1973)",
    x = "Date"
  ) +
  theme_minimal()

#BUBBLE PLOT
# Most basic bubble plot
new %>%
  arrange(desc(Solar.R)) %>%
  ggplot(aes(x=Ozone, y=Temp, size = Solar.R, color=factor(Month))) +
    geom_point(alpha=0.5) +
    scale_size(range = c(.1, 10), name="Solar.R") +
    labs(x="Ozone (ppb)", y="Temperature (°F)", title="Temperature v. Ozone by Month and Solar Radiation")
library(rgl)
Data <- airquality[, c("Ozone", "Solar.R", "Temp", "Month")]
Data <- na.omit(Data)
Data <- as.data.frame(lapply(Data, function(x) as.numeric(as.character(x))))

mycolors <- c('royalblue1', 'darkcyan', 'oldlace', 'magenta', 'aquamarine')
month_colors <- setNames(mycolors, 5:9) # or however many months you have
Data$color <- month_colors[as.character(Data$Month)]

# 3D Plot
plot3d(
  x=Data$Ozone, y=Data$Solar.R, z=Data$Temp,
  col = Data$color,

```

```

type = 's',
radius = 5,
xlab="Ozone", ylab="Solar Radiation", zlab="Temperature",title="3D Graph Temperature v. Ozone v. Solar Radiation"

# To display in an R Markdown document:
rglwidget()
library(ggExtra)

# classic plot :
p <- ggplot(new, aes(x=log(Ozone), y=Solar.R)) +
  geom_point() +
  geom_smooth(method='lm') +
  theme(legend.position="none") + labs(x="log(Ozone)", y="Solar Radiation (Langleys)", title="Solar Radiation vs log(Ozone)")

# with marginal histogram
p1 <- ggMarginal(p, type="histogram")
p1
pred_out <- predict(object = lm(Temp ~ Ozone + factor(Month) + Solar.R, data=airquality),
  newdata = newdata,
  interval = "predict")

# Example data
actual <- new$Temp
predicted <- pred_out[, "fit"]

# Compute Mean Squared Error
mse <- mean((actual - predicted)^2)
mse
rmse <- sqrt(mse)
rmse
pred_out <- predict(object = lm(Temp ~ log(Ozone) + factor(Month) + Solar.R, data=airquality),
  newdata = newdata,
  interval = "predict")

# Example data
actual <- new$Temp
predicted <- pred_out[, "fit"]

# Compute Mean Squared Error
mse <- mean((actual - predicted)^2)
mse
rmse <- sqrt(mse)
rmse

```